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Key Points:

- Validating thermodynamic models against crystallization experiments of arc magmas
- Using thermodynamic models to reduce sparsity of experimentally derived data
- Training neural networks for rapid inference of thermodynamic properties with surrogate models

Supporting Information:

Supporting Information may be found in the online version of this article.

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Validating Thermodynamic Models of Arc-Magma Differentiation and Training Neural Networks for Rapid Thermodynamic Property Inference

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Abstract Magmatic systems are the products of the migration and storage of compositionally evolving magmas within the solid crust. Direct computation of thermodynamic properties during magma transport remains a major challenge in multiphase-transport modeling due to its computational cost. Here, we present a proof-of-concept neural network-based workflow that addresses these challenges. First, we validate thermodynamic databases for modeling arc magma differentiation by comparing the Gibbs Free Energy Minimization (GFEM) software MAGEMin to a compilation of laboratory experiments. Second, we use MAGEMin to generate a data set spanning the entire temperature, pressure, and compositional space relevant to arc magmas. This data set is then used to train a neural network for rapid inference of thermodynamic properties. Results indicate that MAGEMin-predicted liquid compositions and densities generally agree well with those determined experimentally. The accuracy of mineral phase abundance predictions depends on the starting composition. Notably, the ability to predict amphibole stability by the thermodynamic model is better compared to other thermodynamic packages. The network demonstrates excellent performance and accuracy in predicting the thermodynamic properties of arc magmas. Liquid composition and density are predicted with root-mean-square errors below 1 wt.% and 16 kg m⁻³, respectively. Network estimates are ca. 3,000 times faster on the CPU compared to GFEM calculations and 10,000 data points can be calculated in under 1.0 ms on the graphics processing unit. Our workflow offers an efficient approach to the computational challenge of integrating composition-dependent physical properties into reactive transport algorithms. For operational use, the network should be re-trained on application-specific data sets.

Plain Language Summary Understanding magmatic systems is crucial for predicting explosive volcanic eruptions and for genetic models of mineral resources required by society. Fundamental processes governing these systems include compositional evolution, transport across, and storage of magmas at different depths in the solid crust. These processes act on spatio-temporal scales that are too large to observe or reproduce directly in the laboratory and numerical modeling is a key method to furthering our understanding of magmatic systems. Current numerical models suffer when solving magma's compositional evolution directly together with its transport, mostly because calculating the stable phases during magma evolution is computationally expensive. Here, we address these challenges and present a workflow that involves traditional thermodynamic models and innovative machine learning techniques. First, we validate thermodynamic models of arc magma evolution by comparing them to laboratory experiments. Second, we train a neural network to perform high-dimensional interpolation of thermodynamic properties of arc magmas over a large composition, temperature, and pressure range relevant to arc settings. The network shows excellent performance and accuracy in predicting the thermodynamic properties of arc magmas.

1. Introduction

Magmatic systems are the complex products of the ascent, storage and compositional evolution of magmas at multiple depths in the crust (Cashman et al., 2017). Such magmas may eventually reach the surface to form hazardous volcanic eruptions, or may stall in the crust and outgas large volumes of fluid to form economically valuable ore deposits. The fate of a magmatic system is a complex interplay between the mechanics of the crust and the chemical evolution of the magma as it cools and differentiates (Bachmann & Huber, 2016; Sparks et al., 2019).

As magma cools and differentiates to more silica-rich compositions, it decreases in melt fraction and residual melts generally become less dense, more viscous and saturate in a H₂O—dominant fluid phase. Geodynamic

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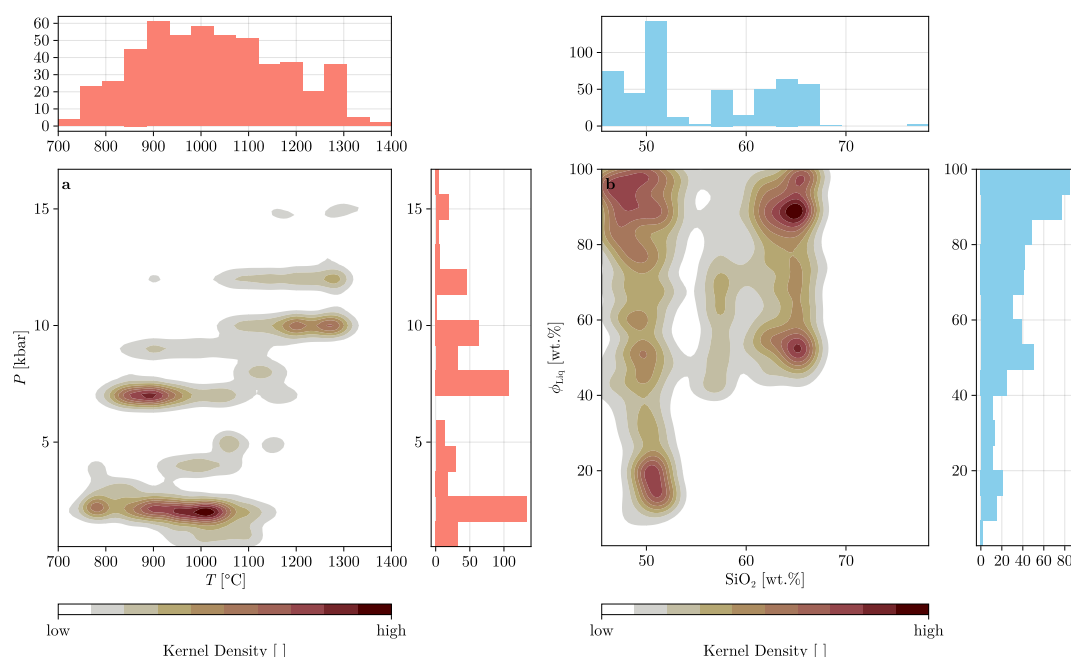


Figure 1. Two-dimensional kernel-density estimates showing (a) the distribution of experiments in temperature–pressure space and (b) the silica content of the starting material versus the liquid fraction at the end of the experiment. The accompanying histograms display how many experiments were conducted at a given value of each variable shown on the axes of (a) and (b). A Gaussian kernel with standard deviations of 25°C and 0.1 kbar was used for smoothing. The data show sampling biases between 950 and 1,100°C, at pressures of 3–4 and 6–7 kbar, and for silica contents near 50 and 65 wt.%.

models of magmatic systems aim to accurately incorporate these changes in chemistry and physics of the magma over time. Typically, such models predict the compositional evolution of magma using experiments performed at equivalent temperature, pressure, and composition (Abdullin et al., 2024; Annen et al., 2006). However, the experimental database sparsely occupies the pressure–temperature–composition (P–T–X) space, with clustering of data sets at 2–3 and 7–10 kbar, and at basaltic and dacitic compositions (Figure 1). Therefore, this approach is less appropriate in cases where magma deviates from the P–T–X space of the calibration data set, and in cases where models span large P–T–X space, such as in transcrustal magmatic systems where mid-crustal magma evolution has been observed as a key staging site in arcs (Glazner & Ussler, 1988; Jagoutz, 2014). Consequently, some studies integrated numerical models with thermodynamic software (Gibbs Free Energy Minimization [GFEM]) such as *Perple_X* (Connolly, 2005), *MELTS* (Ghiorso & Sack, 1995), *Theriak Domino* (de Capitani & Brown, 1987), and most recently, *MAGEMin* (Riel et al., 2022), to predict melt evolution at a wide range of P–T–X conditions (Afonso et al., 2013; Rummel et al., 2018; Rummel, Baumann, & Kaus, 2020).

Direct coupling of geodynamic and thermodynamic models is computationally challenging due to fundamental differences in their parallelization paradigms: while geodynamic solvers are typically optimized for distributed or shared-memory parallelism, thermodynamic calculations remain largely serial. As a result, solving the thermodynamic problem dominates runtime, creating a bottleneck relative to the highly parallelized mechanical solve. The currently most feasible approach is to compute a “lookup table” using a GFEM software (e.g., L. G. Candioti et al., 2021; Gerya et al., 2004; Kaus et al., 2005; Yamato et al., 2007, and others) which works well for constant bulk rock compositions. Modeling the compositional evolution of the system requires pre-computing a large number of phase diagrams and nonlinear interpolation in a ≥ 10 -dimensional space (Rummel, Baumann, & Kaus, 2020) which is non-trivial. Riel et al. (2022) introduced a new thermodynamic software package called *MAGEMin* which is more suitable for coupling with geodynamic codes than existing software, because it is parallelized (scalable) and therefore fast. *MAGEMin* is a C-Library and shipped with a Julia wrapper allowing for almost seamless integration into geodynamic codes independent of the computational platform used and it implements a range of thermodynamic databases.

While a single point MAGEMin pseudo-section evaluation can be executed in the order of 100–200 ms depending on especially the water content of the system (Riel et al., 2022), such execution times are still significantly larger compared to solving the mechanical part of the problem in geodynamic algorithms which operate over thousands of datapoints at any time step (Kerswell et al., 2024). In addition to parallelization, MAGEMin offers the possibility to use initial guesses to accelerate the minimization, yielding an additional speed-up of up to one order of magnitude in best cases. Yet, execution times ≤ 1 ms necessary for efficient coupling to geodynamic codes still seem to be unreachable when solving the complex thermodynamic system directly.

To overcome this barrier, more and more studies have suggested and implemented neural network and machine learning techniques that emulate phase equilibria calculations (De Lucia & Kühn, 2021; Demirel et al., 2023; Guérillot & Bruyelle, 2020; Jatnieks et al., 2016; Kerswell et al., 2024; Leal et al., 2020). While these studies consistently report 10–1,000 fold acceleration by using a so-called surrogate model—a model that replaces another for a given task, in this case, estimating thermodynamic properties—compared to traditional geochemical solvers, the choice of the training data remains a major challenge and crucially impacts on the accuracy of the surrogate model. Especially in time-evolving simulations, adding physical constraints during the training procedure is often required to reduce accumulating errors of surrogate-model predictions (Silva et al., 2025). These challenges need to be addressed carefully when extending such techniques used to model reactive fluid transport in the subsurface to melt migration in transcrustal magmatic systems.

Our understanding of the evolution of arc magmas is largely constrained by laboratory experiments (Müntener & Ulmer, 2018). Figure 1 shows a kernel density plot of P-T-X conditions of more than 400 experiments on arc magmas visualized over the P-T ranges typical of volcanic arcs. Largely due to the operating conditions of the experimental devices (cold seal pressure vessels, internally heated pressure vessels, piston cylinder), most of these experiments cluster in certain regions of the P-T-X space and therefore do not cover all of the possible conditions an arc magma may be stored at. Hence, using only the experimental data to train a neural network is insufficient to guarantee accurate predictions. The experimentally covered area in Figure 1 can be complemented by generating a large number of phase diagrams for realistic arc magma compositions with a thermodynamic software. This method ensures sufficient coverage and a more comprehensive data set the network can be trained on.

In this study, we present a proof-of-concept neural network-based surrogate model that emulates the phase equilibria calculations for a broad range of arc-magma compositions. The goal is to generate a fast, high-dimensional nonlinear interpolator that is aware of the complex relationships between thermodynamic input and output quantities of arc magmas. To this end, we first demonstrate that the thermodynamic database of Holland et al. (2018), implemented by MAGEMin, can accurately reproduce the compositional evolution of arc magmas by comparison with experiments. Based on this validation, we use MAGEMin to generate a compositional database that spans the P-T-X space relevant to the evolution of arc magmas in the crust. We develop a neural network architecture that can be used to rapidly infer these compositions offering high potential to be coupled to geodynamic codes. This proof of concept demonstrates the feasibility of the approach. For operational use, the choice of (a) data on which the network is trained and (b) the thermodynamic properties it predicts should be made according to the flow-regime that is predicted by the geodynamic model.

2. Methods

2.1. Experimental Compilation

Benchmarking of the thermodynamic models was performed by comparing experiments with outputs of the thermodynamic software for the same temperature, pressure and bulk composition. To stay consistent, we adopt the MAGEMin nomenclature for three-phase aggregates. Hereafter, we will use the terms liquid, fluid, and solid for the melt, magmatic volatile phase, and the stable mineral phases, respectively. To produce the data set upon which the applicability of MAGEMin to arc magma evolution could be tested, a previously published experimental compilation was adapted (Weber & Blundy, 2024). This data set contains the experimental conditions (temperature, pressure, oxygen fugacity, run duration), the composition and fraction of the liquid at the end of the experiment as well as the fraction of each stable mineral phase in the solid. We include the experiments of Johnston (1986), Baker and Eggler (1987), Kelemen et al. (1990), Sisson and Grove (1993a, 1993b), Blatter and Carmichael (2001), Barclay (2004), Villiger (2004), Kagi (2005), Sisson et al. (2005), Müntener and Ulmer (2006), Botcharnikov et al. (2008), Blatter et al. (2013), Nandedkar et al. (2014), Blatter et al. (2017), Marxer and Ulmer (2019), and

Blatter et al. (2023) (see Figure 1 for experimental data density maps). Additional data that were compiled are the major element composition of the starting material (a required input for MAGEMin) and the measured abundance of each mineral phase at the end of the experiment. For experiments which were fluid-saturated, the initial H_2O content of the starting material was set as the water solubility calculated using the Python package Vesical (Iacovino et al., 2021) using the water solubility model of MagmaSat (Ghiorso & Gualda, 2015). To compare experimental liquid densities with those computed by MAGEMin, the density of experimental liquids was calculated using the Python package *DensityX* (Iacovino & Till, 2019). The full experimental compilation consists of 484 experiments and is available as “MineRS_MagmaDB_merged_v2_H2Ocalc.csv” in the related GitHub repository (see Availability Statement section).

2.2. Gibbs Energy Minimization Using MAGEMin

2.2.1. MAGEMin Setup

Gibbs Free Energy minimizations in this study were performed using the Julia interface of MAGEMin (Riel et al., 2022), MAGEMin_C.jl (v1.8.1 for experimental comparison and v1.9.9 for all the network-related data). The software uses linear programming, extended Partitioning Gibbs Energy and gradient-based local minimization to compute the stable mineral phase and melt composition for a given temperature, pressure and bulk composition. The igneous thermodynamic database of Holland et al. (2018), which was later updated by Green et al. (2025), was selected. Our comparison is complementary to the benchmarking of the igneous anhydrous alkaline database in MAGEMin performed by Weller et al. (2024), which shares some equations of state with the igneous database. In our benchmarking, it is particularly important to test whether hydrous mineral phases (amphibole and biotite) are adequately predicted because some approaches (e.g., MELTS) are not appropriate for hydrous mineral phase equilibria (Gualda et al., 2012). For the experimental comparison, we included all available solution models and pure phases from the igneous database, then renormalized the liquid major oxide concentrations to anhydrous compositions for consistency. However, when generating the data sets for training and testing the neural network, we excluded the “fl” (fluid) phase from the pure phase models. This exclusion prevents major oxide partitioning into the fluid phase and assumes the free fluid phase is pure water—a simplification justified by the current limitations of MAGEMin in modeling detailed fluid chemistry. Depending on the experiment that was being tested, the buffer function of MAGEMin was activated to buffer the $f\text{O}_2$ at the same buffer (e.g., nickel-nickel oxide) and delta value (e.g., +1 log units) reported in the experiment. In order to activate the buffer, excess O is required in the bulk composition (Weller et al., 2024), and 4 mol% was added to each composition as this was found to ensure the buffer was always present. A point-wise minimization was performed on each experimental starting composition, pressure and temperature, and the liquid composition and phase abundances were compared with those reported by the experiment.

2.2.2. Generation of Training Data Set of Arc Magma Compositions

To generate the training data set of arc magma thermodynamic properties in the crust, a P-T-X parameter space must be constructed. For temperature, a range of 650–1,300°C was used to cover the maximum range from solidus to supra-liquidus conditions. For pressure, a range of 0.1–10 kbar was used to cover the range from the surface to the base of the crust in a continental arc. The compositional range must be a multivariate distribution of arc magma compositions that can be explored. We selected nine major oxides relevant to the igneous database of Holland et al. (2018): SiO_2 , Al_2O_3 , CaO , MgO , FeO_T , Na_2O , K_2O , TiO_2 , and H_2O . Cr_2O_3 was excluded for two reasons: (a) its abundance in arc magmas is negligible, and (b) it is often unreported in laboratory experiments. However, further reducing dimensionality by omitting TiO_2 —in addition to Cr_2O_3 —is not advisable, as it directly influences spinel and ilmenite stability and, consequently, the calculated densities. Capturing the covariance of these compositional data is important, because treating each major element as separate univariate distributions may lead to unrealistic compositions (e.g., high SiO_2 and high MgO). To derive an initial compositional space to be sampled, a database of whole-rock major element compositions was compiled (Nathwani et al., 2022) which was adapted and filtered from the GEOROC database (Lehnert et al., 2000). Volcanic rocks were used as opposed to plutonic rocks, since the former more closely represent liquid compositions rather than accumulated crystals. Because this compositional database is likely biased toward the more evolved compositions that reach the surface, it was supplemented with a whole-rock database of primitive arc rocks (Schmidt & Jagoutz, 2017). Compositions that had any major oxide that was not reported were not included

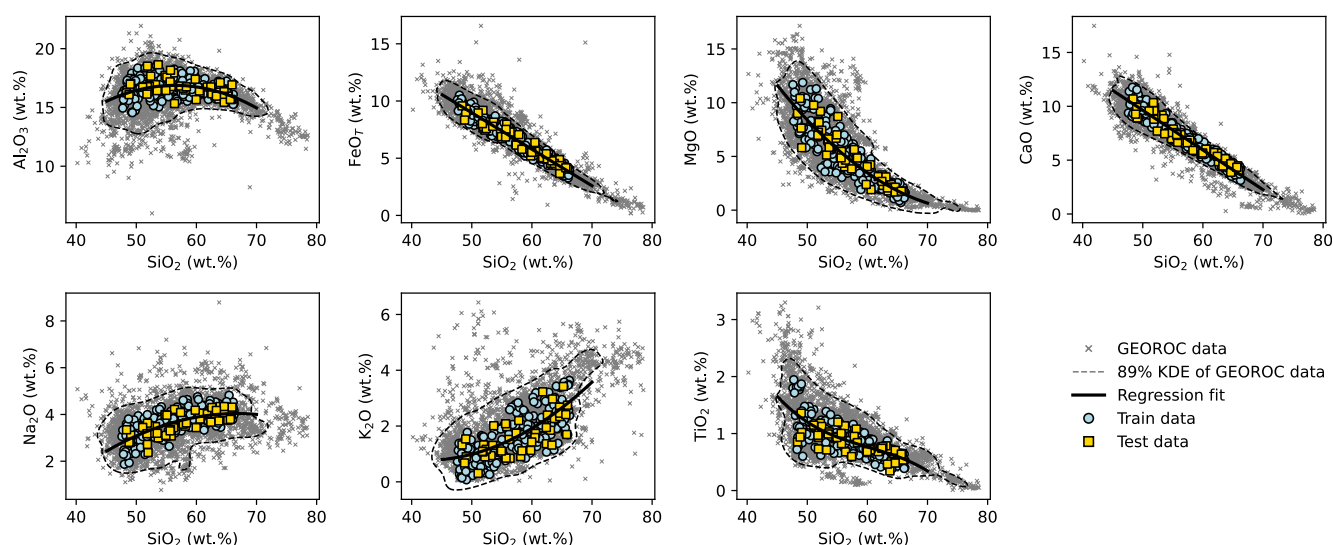


Figure 2. Harker diagrams of oxides (a–g) used for data generation with MAGEMin. The compositions are taken from the GEOROC database. Blue circle markers show the training compositions, yellow square markers the testing compositions. The black solid line represents the general data trend and the dashed contour the border within which 89% of the data points fall.

in the final database. The total whole-rock geochemical database consists of 4,516 compositions (960 from the primitive arc magma compilation and 3,556 from the GEOROC database).

The initial compositional database exhibits very high variance and a large number of outliers. Our goal is to sample 100 compositions that best define the general compositional variance of arc magmas with the lowest number of data points, and without oversampling low probability regions. We therefore designed a workflow to generate synthetic compositions to train and test the neural network (Figure 2). First, outliers are filtered from the compositional space by discarding data that lie outside of the 89% probability mass of the kernel density estimation. This threshold was found to isolate the main cluster of data points. Above that threshold several smaller clusters appeared outside the main data trend. Then, a regression of each major element is performed against SiO_2 , using a least squares polynomial fit: Al_2O_3 (2), FeO_7 (1), MgO (2), CaO (1), Na_2O (2), K_2O (2), and TiO_2 (3), where the number in parentheses is the order of the polynomial function. These regressions are used to calculate the major element composition of randomly drawn SiO_2 contents between 47 and 66 wt.%. To incorporate some variance into these compositions, Gaussian noise is added to each synthetic composition according to the standard deviation for each major element within an allocated SiO_2 bin. Lastly, due to the presence of rare outliers, the final data set was filtered once more to ensure that all points lie within the 89% probability contour of the initial GEOROC database. A total of 100 synthetic compositions were generated by this approach, which were used to generate 100 phase diagrams using MAGEMin. To include the impact of water we added a constant value of 4 wt.% H_2O to the initial compositions. Although a range of water contents are found in arc magmas, we found that fully accounting for the possible variability in initial H_2O concentrations drastically increased the size of our data set. Any future studies using such surrogate models should constrain the P-T-X space that they need to generalize for their specific problem case so that the surrogate model can be trained and optimized within realistic time frames.

Here, we benefit from two main computational advantages of MAGEMin compared to other GFEM software packages: (a) MAGEMin supports multi-threading which allows for parallel execution of minimization, and (b) MAGEMin features adaptive mesh refinement. This numerical technique allows for high resolution phase diagram-computations while keeping the data sets small enough to train on a single graphics processing unit (GPU) device. We use five initial grid levels with two additional refinement levels to generate the phase diagrams. The final data set we used for training the network consists of ca. 980,000 equilibrium assemblages.

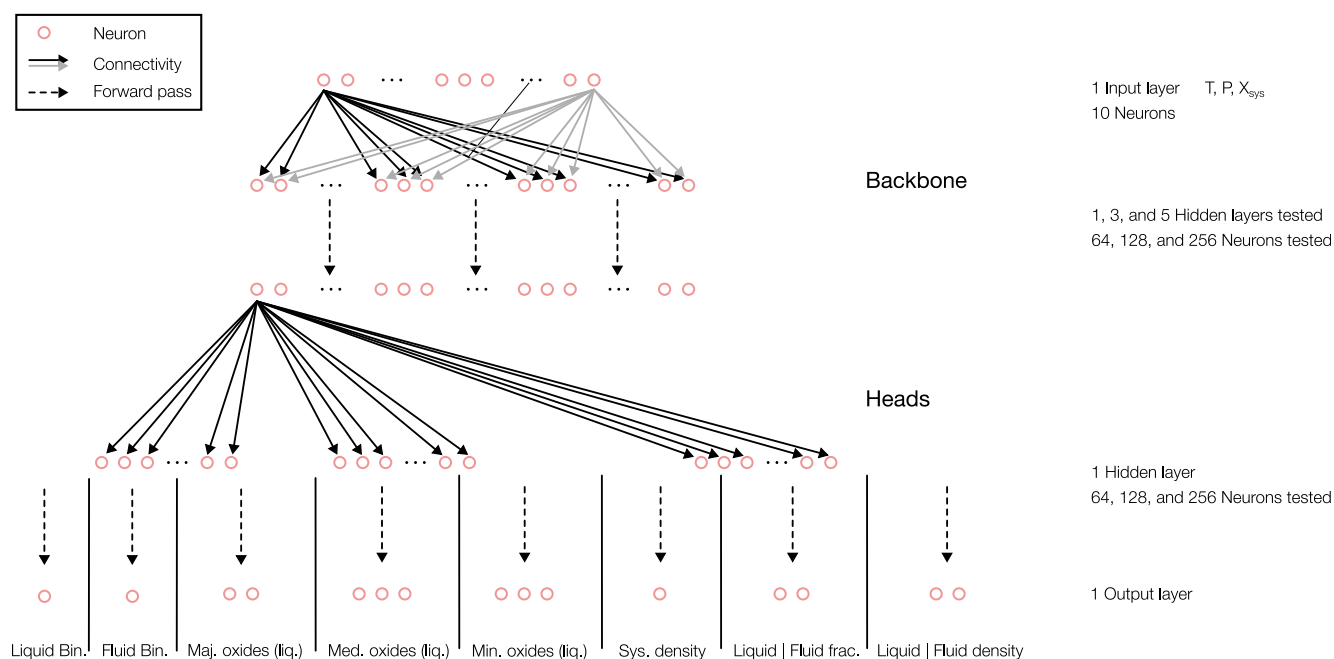


Figure 3. Architecture of the multi-task learning multi-layer perceptron. Salmon-colored circles represent neurons, black, and gray solid arrows show neuron connectivity, and black dashed arrows the direction of the forward pass.

2.3. Multi-Task Learning Multi-Layer Perceptron

We mainly use two Julia packages to configure and train the neural network: Lux.jl (Pal, 2023, v1.13.2) and Optimizers.jl (v0.4.6) which is part of the Flux.jl package (Innes et al., 2018). Lux provides the main framework for our machine learning algorithm. It is shipped with constructors that build the network's architecture, initialize arrays on both the CPU and the GPU (via LuxCUDA.jl, v0.3.3), define loss functions, and compute the gradients with automatic differentiation packages such as Zygote.jl (v0.7.7) (Innes et al., 2019).

2.3.1. Network Architecture

The main goal of the neural network is to interpolate thermodynamic properties of the system within a ten-dimensional space. Hence, the neural network acts as a multi-dimensional, non-linear regression tool. As we are considering three-phase systems (solid, liquid, and fluid) we need at least two classifiers that predict whether liquid and fluid coexist with the solid. A suitable type of neural network that can be applied here are multi-task learning multi-layer perceptrons (MTL-MLP, e.g., Crawshaw, 2020). These networks consist of a shared backbone that expands and extracts the relationship between input and output layers and different heads that are responsible for making predictions for the thermodynamic output parameters. A graphic representation of the architecture used in this study is shown in Figure 3. Analogous to MAGEMin's input variables, the surrogate model will take temperature, pressure, and the system concentration of SiO_2 , CaO , Al_2O_3 , MgO , FeO_T (FeO hereafter), $\text{Na}_2\text{O} + \text{K}_2\text{O}$, TiO_2 , and H_2O as input variables. Combining $\text{Na}_2\text{O} + \text{K}_2\text{O}$ reduces the dimensionality of the input parameter space. This simplification is justified as the combined values mirror the SiO_2 trends of individual species, ensuring the primary geochemical signal is preserved (see Figure S6 in Supporting Information S1). Note that we only combine alkalis for the network. In the context of the experimental comparison and for generating the training data with MAGEMin, Na_2O and K_2O are considered separately. To improve the performance and robustness of the neural network, we applied a logarithmic transformation to the input parameters (pressure, CaO , MgO , $\text{Na}_2\text{O} + \text{K}_2\text{O}$, and TiO_2) and the output parameters (CaO , MgO , FeO , and TiO_2 , concentrations in the liquid). Prior to training, all input parameters were further standardized using Z-score normalization, ensuring that the training procedure operated entirely on scaled values. This preprocessing step helps mitigate the impact of varying parameter magnitudes and distributions, facilitating more effective convergence behavior.

Table 1
Tested Hyperparameters and Their Values

Hyperparameters	Values
Batch size	[1,024, 4,096, 16,384]
No. layers	[1, 3, 5]
No. neurons	[64, 128, 256]

To reduce the dimensionality of the output parameter space, we choose to train the network only on liquid, fluid, and system properties. Any intensive thermodynamic property of the solid, A_{Sol} can be computed from the predicted quantities in post-processing enforcing the mass balance constraint:

$$A_{\text{Sol}} = \frac{A_{\text{Sys}} - A_{\text{Liq}}\phi_{\text{Liq}} - A_{\text{Flu}}\phi_{\text{Flu}}}{1.0 - \phi_{\text{Liq}} - \phi_{\text{Flu}}}, \quad (1)$$

where, A_{Sys} , A_{Liq} , and A_{Flu} are the system, liquid and fluid property, ϕ_{Liq} and ϕ_{Flu} are the liquid and fluid phase fractions. While this approach significantly reduces the complexity of the neural network, it has a computational limit: At high mobile phase (combined liquid and fluid) fractions, the denominator in Equation 1 approaches 0 for which this equation is not valid. We observed consistent solid density calculations for mobile phase fractions as high as 80 wt.% (see also Text S5 and Figure S5 in Supporting Information S1). In scenarios characterized by substantial proportions of the data set exhibiting elevated mobile phase fractions, it is recommended to directly train the network on liquid, fluid, and solid phase properties.

Here, the network is trained to (a) classify the presence or absence of liquid and fluid, to predict (b) the concentration of the above-mentioned major oxides (split into three separate heads according to their abundance) in the liquid, (c) the system density, (d) the liquid and fluid fractions, and (e) the liquid and fluid densities. We chose these parameters because they are required by most multi-phase geodynamic algorithms and, compared to other quantities such as heat capacity, the chosen parameters have higher variability in their numerical range. The 10 input features are passed through the backbone—a sequence of hidden layers each consisting of a defined number of neurons. From the deepest layer inside the backbone the information is branched out into the eight different heads.

2.3.2. Training and Validation Procedure

The number of layers and the number of neurons per layer are referred to as hyperparameters of the network. They control the network's capacity to capture patterns in the data but are not directly learned during training. Instead, their optimal values are found through a systematic parameter study (see Table 1). For each parameter combination, we apply a cross-validation procedure to assess the surrogate model performance and robustness. Specifically, we use a five-fold cross-validation, meaning that the training data are divided into five different splits, each consisting of an 80% subset for training and a 20% subset for validation. During each of the five iterations, the 80% subset is used to update the network parameters by minimizing the loss function, while the remaining 20% subset tests how well the network generalizes to unseen data. This approach not only identifies the best hyperparameter settings but also helps detect overfitting and evaluate the network's stability and reproducibility (Morin & Davis, 2017; Mosier, 1951).

The training itself is an iterative optimization process that minimizes the mismatch between the surrogate-model predictions and the training data. We fix the number of iterations—referred to as epochs—to 500, meaning that the entire data set is passed through the network 500 times. We found this number to be sufficient, as the losses plateau earlier for our data set. Training is performed on an NVIDIA RTX-A6000 GPU with 48 GB of memory to accelerate computation. We apply batch training, where the training data are divided into smaller groups, or batches. This approach balances computational efficiency and accuracy: larger batches reduce training time but may limit the network's ability to generalize to new data, whereas smaller batches introduce more stochastic variation that can improve generalization (e.g., Keskar et al., 2017). Together with the number of layers and neurons per layer, the batch size forms the third key hyperparameter optimized in our cross-validation study. Our hyperparameter search prioritized architecture-related parameters (depth, width, batch size), because preliminary tests indicated these had the greatest impact on performance. While future work could further explore learning rate schedules and optimization parameters, we recommend conducting a more comprehensive hyperparameter search tailored to the specific data set for operational applications.

To capture the complex, nonlinear relationships between the input parameters and the thermodynamic properties of the liquid, the fluid, and the total system, we use the following layer, l , architecture:

$$\hat{\mathbf{y}}^{(l)} = \text{act}(\hat{\mathbf{x}}^{(l)}) \quad (2)$$

$$\hat{\mathbf{x}}^{(l)} = \mathbf{W}^{(l)}\hat{\mathbf{y}}^{(l-1)} + \mathbf{b}^{(l)}. \quad (3)$$

In Equation 2, $\text{act}()$ denotes the nonlinear activation function, which introduces nonlinearity and enables the network to approximate complex relationships. We use the $\tanh$ activation function because it is well suited to the nonlinear character of the input–output relationship in our data. Equation 3 describes how, in each layer, the outputs of the previous layer are linearly combined using a weight matrix and a bias term. In the first layer, $\hat{\mathbf{y}}^0 \in \mathbb{R}^{d_{(0)} \times N_{\text{pts}}}$ in Equation 3 corresponds to the actual input data, while in all subsequent layers, the inputs are the outputs of the preceding layer. The weight matrix, $\mathbf{W} \in \mathbb{R}^{d_{(l)} \times d_{(l-1)}}$ and bias vector $\mathbf{b} \in \mathbb{R}^{d_{(l)} \times N_{\text{pts}}}$ are the adjustable parameters of the network. They are initialized with random numbers drawn from a normal distribution and zeros, respectively (Glorot & Bengio, 2010). The output of each layer, $\hat{\mathbf{y}}^{(l)}$ is passed forward through the network until the final predictions are obtained.

Once the forward pass is completed, the surrogate model predictions are compared to the known training data using loss functions, which quantify how well the surrogate performs. For the classifier heads, which are responsible for predicting whether a phase (liquid or fluid) is present, we use the binary cross-entropy loss:

$$\mathcal{L}_{\text{BCE}} = \sum_{i=1}^n (-\tilde{y}_i \log(\hat{y}_i + \epsilon) - (1 - \tilde{y}_i) \log(1 - \hat{y}_i + \epsilon)) / n \quad (4)$$

$$\tilde{y}_i = (1 - \alpha)y_i + 0.5\alpha, \quad (5)$$

where α denotes a label smoothing parameter that prevents overconfidence in classification. For the regression heads, which predict continuous quantities such as thermodynamic properties, we use the Huber loss:

$$\mathcal{L}_{\text{HUB}} = \begin{cases} 0.5|y - \hat{y}|^2 & , |y - \hat{y}| \leq \delta \\ \delta(|y - \hat{y}| - 0.5\delta) & , |y - \hat{y}| > \delta. \end{cases} \quad (6)$$

This loss combines the advantages of mean squared and absolute errors, being less sensitive to outliers while maintaining stability during optimization. The total loss, $\mathcal{L}_{\text{tot}}$, is computed as the sum of all head losses.

To avoid computing regression losses when a phase is absent, we multiply both the predicted and reference values by a binary mask. This ensures that the Huber loss is zero for samples without a liquid or fluid phase, focusing the optimization only on relevant data. After loss evaluation, we compute the gradient of the total loss with respect to all network parameters and update them iteratively using dedicated optimization algorithms. The backbone and classifier heads are optimized using AdamW—a variant of the Adam algorithm (Kingma & Ba, 2017) with fixed weight-decay regularization—implemented in the *Optimizers.jl* package. The regression heads use the standard Adam optimizer from the same library.

The learning rate, η , controls the magnitude of parameter updates and is kept constant at $\eta = 10^{-3}$ for the first 50 epochs. It is then increased to $\eta = 10^{-2}$ and subsequently adapted using cosine annealing (Liu, 2022, e.g.), which gradually lowers the learning rate as training progresses

$$\eta_{\text{curr}} = \eta_{\text{min}} + 0.5(\eta_{\text{max}} - \eta_{\text{min}}) \left(1 + \cos\left(\frac{T_{\text{curr}}}{T_{\text{freq}}}\pi\right) \right), \quad (7)$$

where η_{curr} is the current learning rate, $\eta_{\text{min}} = 10^{-3}$ and $\eta_{\text{max}} = 10^{-2}$ define its range, T_{curr} is the current epoch number, and T_{freq} specifies how often the learning rate resets. Cosine annealing helps to converge smoothly by reducing the learning rate toward the end of training. Learning rate scheduling avoids local minima, improves generalization, and enhances the convergence speed. During cross-validation, we disable learning rate resets by setting $T_{\text{freq}} = 500$. After cross-validation, the final network is trained on the full data set for 1,000 epochs, resetting the learning rate every 250 epochs after the warm-up phase.

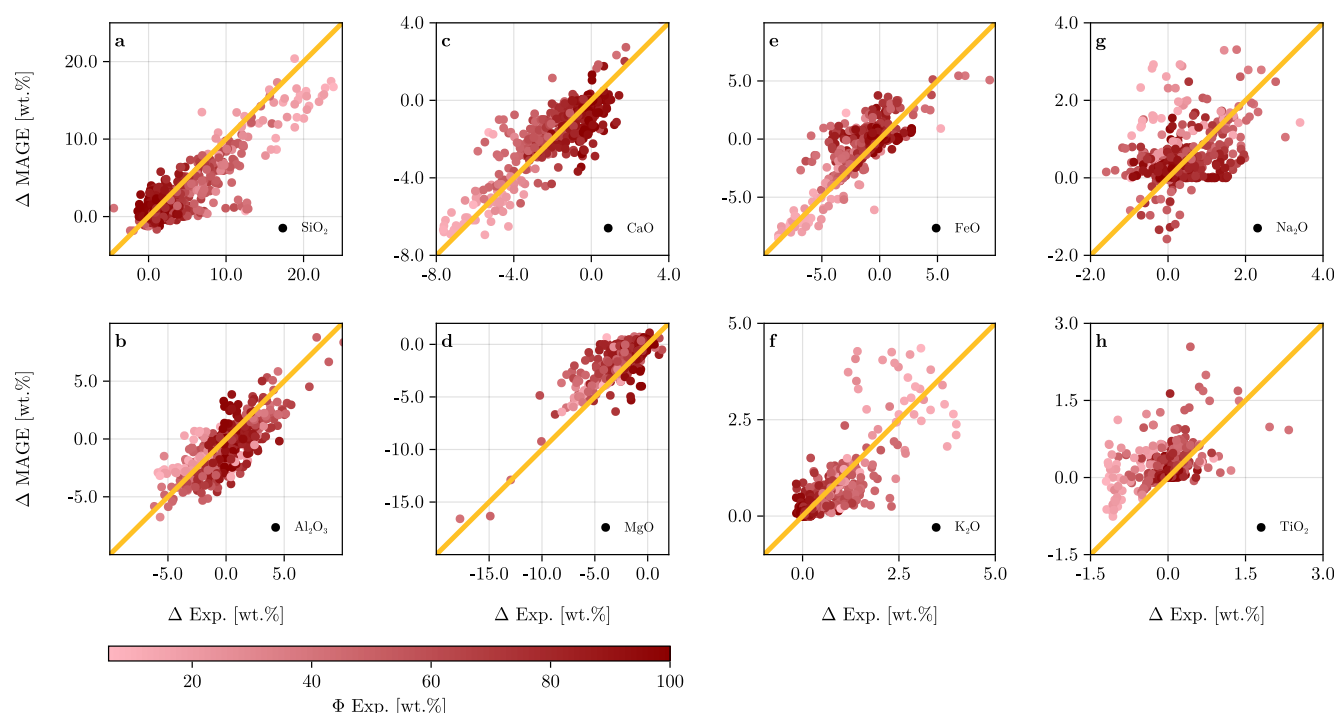


Figure 4. Difference between final liquid and starting material composition for major oxides. The horizontal axis shows the concentrations reported in the experiments, the vertical axis shows the concentrations predicted by MAGEMin. The solid golden line indicates unity, that is MAGEMin and experimental concentrations are identical.

Although the learning rate and decay constants are often treated as additional hyperparameters, their influence is strongly data-dependent, particularly on data scaling (see Text S1, Figures S1a and S1b in Supporting Information S1). Initial tests showed their impact to be secondary compared to the overall network architecture. To maintain focus and reduce computational cost, we therefore excluded these parameters from the cross-validation study.

After validating the neural network and selecting the optimal hyperparameters, we performed an independent test to evaluate how well the surrogate model generalizes to new data. This testing uses compositions that were not included in the training process. To ensure a fair assessment, these test compositions must share the same multivariate distribution as the training data allowing the surrogate to evaluate within its learned parameter space while avoiding overlap with the training database of whole-rock compositions. To create the test set, we generated 75 new compositions using the same regression approach and Gaussian noise propagation methods employed during training data generation.

3. Results and Discussion

3.1. The Applicability of Thermodynamic Models for Predicting Phase Equilibria in Arc Settings

The aim of the comparison of MAGEMin and experiments at equivalent P-T-X is to test whether changes in liquid and solid composition observed in experiments can be accurately reproduced by MAGEMin (and the assigned thermodynamic database). Instead of comparing the output liquid compositions of the Gibbs energy minimization to the experimental liquids, we calculate the weight-percent difference between the starting composition and the liquid for each major oxide (Figure 4). This allows us to evaluate the ability of the thermodynamic model to predict accurate melt evolution with cooling, rather than its ability to reproduce the starting composition at near-liquidus conditions. Although the data distribution is scattered, the general trend of most oxides is linear indicating a reasonable fit between MAGEMin predictions and experiments. The distributions further indicate that accordance between MAGEMin liquid oxide predictions and laboratory measurements depends on the starting liquid compositions, that is for some compositions the prediction error is very low, whereas for others the error can be quite large.

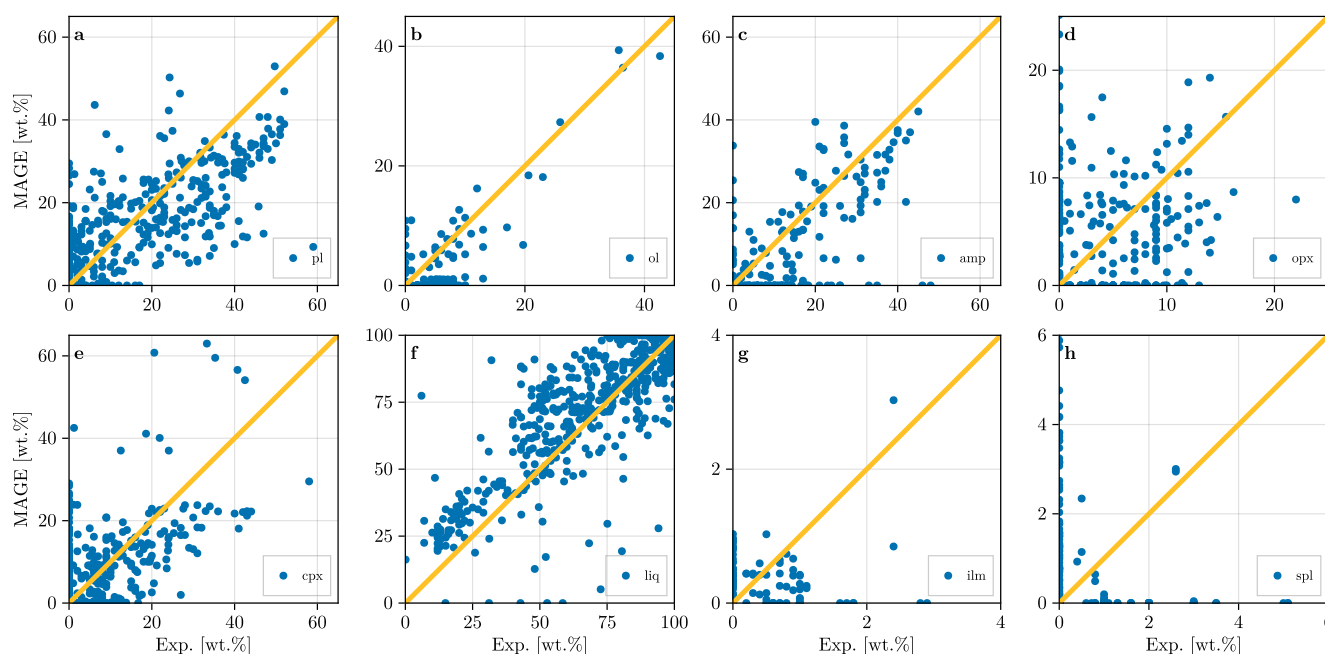


Figure 5. Comparison of phase abundances: ol, olivine; opx, orthopyroxene; cpx, clinopyroxene; pl, feldspar; amp, amphibole; spl, spinel; ilm, ilmenite; liq, liquid. The horizontal axis shows the concentrations reported in the experiments, the vertical axis shows the concentrations predicted by MAGEMin. The solid golden line indicates unity, that is MAGEMin and experimental concentrations are identical.

In detail, MAGEMin predictions of SiO_2 , CaO , Al_2O_3 , and MgO in liquid compare reasonably well to experimentally determined values (Figures 4a–4d). This reflects the accuracy of the thermodynamic model to predict olivine, feldspar and amphibole stability (Figures 5a–5c). In contrast, the poor ability of the thermodynamic models to reproduce changes in Na_2O and K_2O (Figures 4f and 4g) are likely related to the very high mismatch between orthopyroxene and clinopyroxene abundances of the thermodynamic model and experiments (Figures 5d and 5e). The slight over-prediction of FeO (Figure 4e) and strong over-prediction of TiO_2 (Figure 4h) reflects the mismatch between ilmenite and spinel abundances between the models and experiments (Figures 5g and 5h). The lack of consistency between ilmenite and spinel predicted is unsurprising, given that they have a very small contribution to the Gibbs free energy of the assemblage. Additionally, the stability of spinel is very dependent on liquid Cr which is often not reported in laboratory experiments and was—as mentioned above—neglected in this work. The relatively strong performance of the thermodynamic model to predict amphibole stability in hydrous systems (Figure 5c), is particularly significant as it addresses a known limitation of other thermodynamic packages (Gualda et al., 2012).

3.2. Benchmarking Arc Magma Differentiation at Different Pressures

For a more detailed comparison of thermodynamic models to reproduce phase equilibria for arc settings, we used the experiments of Blatter et al. (2013, 2023) which report crystallization experiments of a hydrous basalt at six different pressures appropriate for the mid to lower crust (4.0–16.7 kbar). We found a particularly strong performance of the thermodynamic models in predicting the phase assemblages (Figure 6) and liquid compositions (Figure S2b in Supporting Information S1) of the experiments.

At 4 and 9 kbar pressure, MAGEMin correctly predicts the presence of amphibole (Figures 6a–6d) which is a crucial phase in hydrous magmatic systems, especially in the lower crust. This is an important observation given that other thermodynamic models struggle to predict the stability of amphibole (Gualda et al., 2012). Olivine is not predicted to be stable at 4 and 9 kbar and instead orthopyroxene is predicted to crystallize (Figures 6a–6d). However, olivine is predicted to be stable at 3 kbar (Figure S2a in Supporting Information S1) and thus its stability is within uncertainty, given that the phase boundaries predicted by thermodynamic models have uncertainties as high as 50°C–100°C and 1–2 kbar (Green et al., 2016). Furthermore, Blatter et al. (2023) report uncertainties in their estimated phase abundances of 10%, which can be up to 50% for accessory phases. MAGEMin also predicts

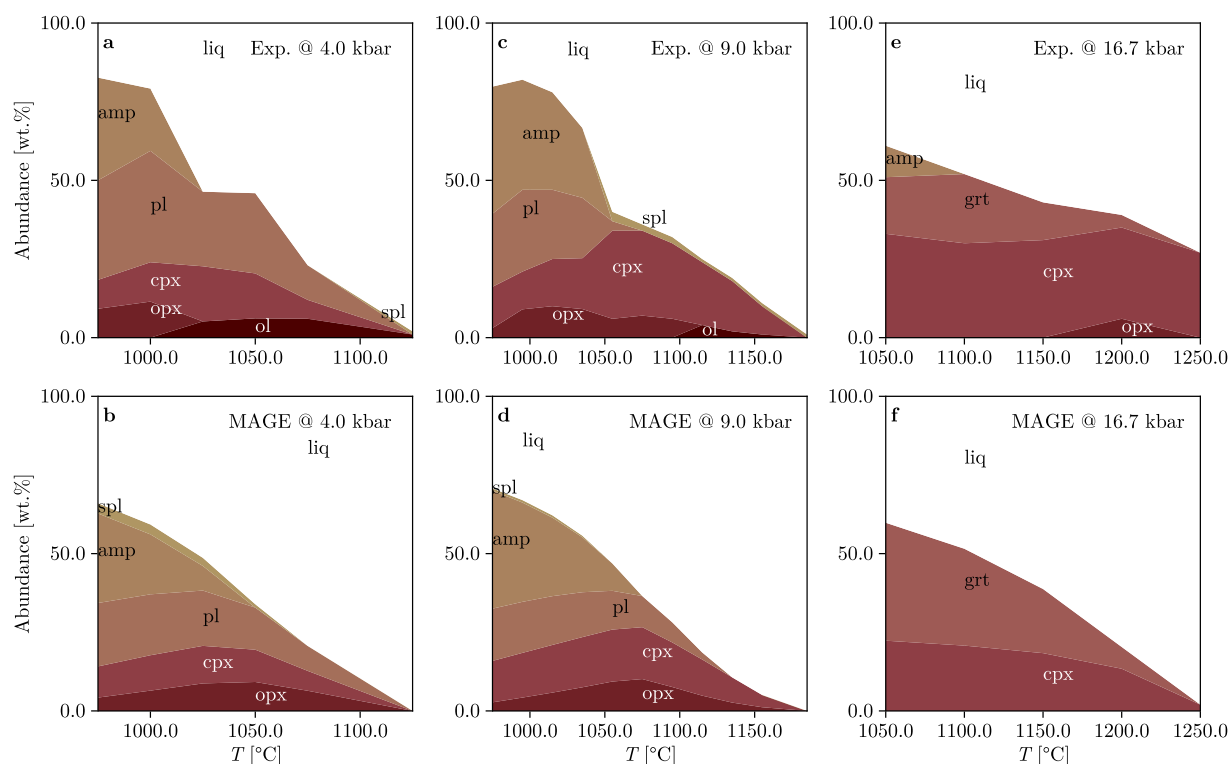


Figure 6. Phase abundance comparison between MAGEMin and the experiments by Blatter et al. (2013, 2023) at different pressures, where ol, olivine; opx, orthopyroxene; cpx, clinopyroxene; grt, garnet; pl, feldspar; amp, amphibole; spl, spinel; liq, liquid. The horizontal axis shows temperature and the vertical axis the mineral abundance.

spinel stability at 4 kbar which is not observed in the experiments at the same temperature. At 16.7 kbar pressure, MAGEMin does not predict the stability of orthopyroxene and amphibole, which is in contrast to the experiments (Figures 6e and 6f). Plagioclase is predicted to be stable at higher temperatures compared to the experiments at 9 kbar. In all of the tested pressure and temperature values, the liquid fraction is consistent between MAGEMin predictions and experimentally determined values (white phase in Figure 6).

3.3. Predicting Physical Properties of Hydrous and Anhydrous Magmas

We find good agreement between liquid fractions predicted by thermodynamic models and experiments (Figure 5f). There is a general bias of MAGEMin to predict higher melt fractions than experiments (Figure 5f). This can also be observed by a complementary bias of MAGEMin to predict slightly lower abundances of some phases, predominantly plagioclase but also clinopyroxene (Figures 5a and 5e). The liquid fraction present in a magma at a given temperature is very sensitive to dissolved H_2O , with this effect being stronger the less H_2O is dissolved (Holtz & Johannes, 1994). Experiments indicate that an additional 1 wt.% dissolved H_2O will increase the melt fraction by approximately 15% (Whitney, 1988).

For some experiments, we find exceptional consistency between temperature–liquid fraction curves outputted by MAGEMin, such as for the hydrous crystallization experiments of a tonalitic magma at 2 kbar by Marxer and Ulmer (2019) (Figure 7a). These authors also compared their experimental temperature–liquid fraction curves with MELTS which indicated good agreement, though we note that the cubic temperature–liquid fraction curve appears to be better captured by the Holland et al. (2018) model than by MELTS (Marxer & Ulmer, 2019). The discrepancy of liquid fractions between MAGEMin and experiments appears to be highest for nominally anhydrous experiments. For example, the nominally anhydrous crystallization experiments of Villiger (2004) have liquid fractions which can be over 50% higher at a given temperature than those predicted by MAGEMin (dark blue solid line with triangle markers in Figure 7b). Rerunning the MAGEMin simulations with 0.5 wt.% H_2O brings consistency between the modeled liquid fractions and experiments (dark blue dashed line with triangle markers in Figure 7b). This may indicate that a small amount of H_2O was present in these experiments. The

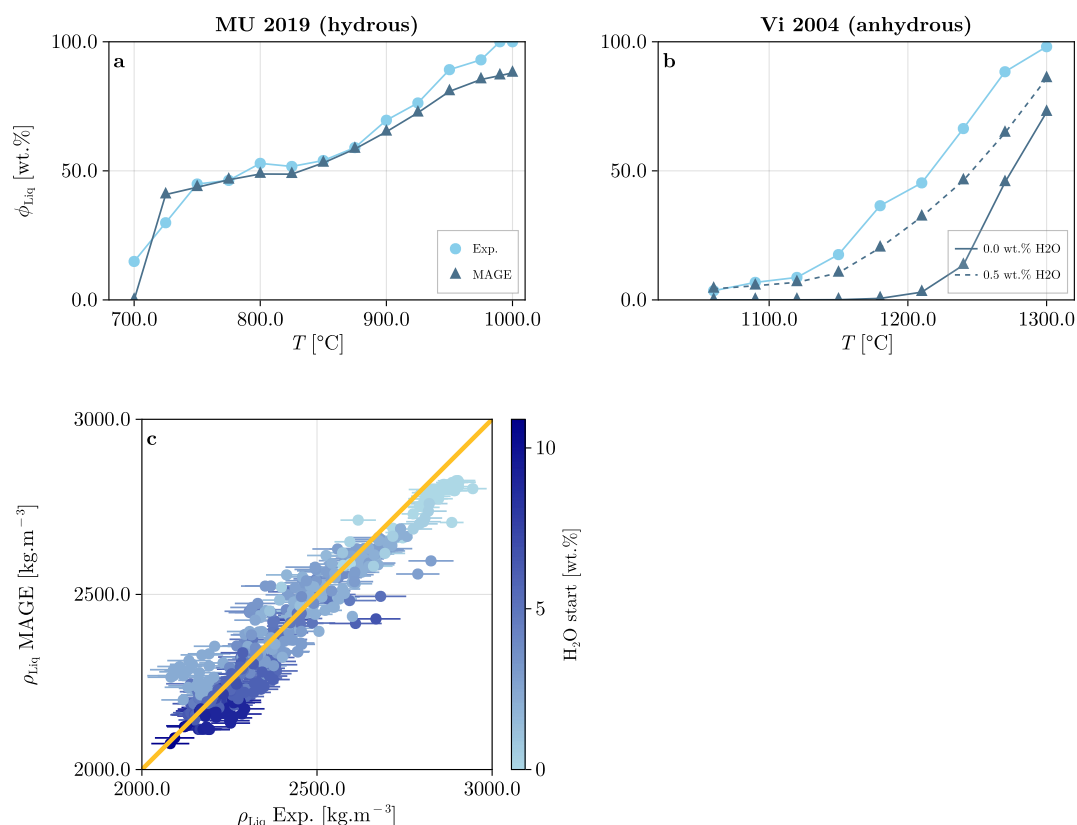


Figure 7. Cumulative liquid fraction for (a) hydrous experiments by Marxer and Ulmer (2019) and (b) anhydrous experiments by Villiger (2004). Light blue-colored circles and solid lines indicate experimental values, dark blue-colored triangles and solid lines indicate MAGEMin calculations. The additional dark blue-colored dashed line with triangles represents a numerical experiment using the exact pressure and temperature conditions as in Villiger (2004) but adding 0.5 wt.% water making the experiment weakly hydrous. (c) Liquid density and its uncertainty calculated by the formula presented in Iacovino and Till (2019). Blue colors indicate the starting concentration of H₂O in wt.%. The solid golden line indicates unity, that is MAGEMin and experimental concentrations are identical.

presence of trace amounts of water in the starting materials of nominally anhydrous experiments is extremely common and can lead to H₂O concentrations in liquids of up to 1 wt.% (Laporte et al., 2004; Médard et al., 2008; Sarafian et al., 2017). In general, we note that the inconsistencies between the thermodynamic model and experiments are within their associated uncertainties.

Overall, density (Figure 7c) predictions of MAGEMin agree well with densities of the experimental liquids calculated by *DensityX* (Iacovino & Till, 2019). Here, the initial water content does not strongly influence the accuracy of predictions.

3.4. Recommendations for Using Thermodynamic Models for Modeling Arc Magma Evolution

Our study confirms the suitability of the thermodynamic database of Holland et al. (2018) and Green et al. (2025), as implemented in MAGEMin (Riel et al., 2022), for modeling arc magma differentiation. The ability of these databases to reproduce experimental phase assemblages is, however, strongly composition-dependent. This dependence likely reflects the compositional range over which the thermodynamic models were originally calibrated. Our analysis found no strong correlations between root-mean-square error (RMSE) and the experimental starting conditions used in MAGEMin. However, we observed moderate correlations with temperature near the liquidus and weak correlations with starting Al₂O₃ (higher accuracy below 13 wt.%) and intermediate TiO₂ concentrations (see Text S7 and Figure S7 in Supporting Information S1). These patterns likely reflect limited crystallization near the liquidus and the thermodynamic database's solution models for phases like spinel, ilmenite, plagioclase, hornblende, and pyroxene. Since there is no correlation between RMSE and starting SiO₂ concentrations, we cannot provide general guidelines for which compositions MAGEMin will predict most

accurately. Consequently, bulk compositions used in modeling should be carefully benchmarked against experimental and natural data to ensure that they fall within the well-constrained calibration space.

For modeling the deep- to mid-crustal evolution of arc basalts, the starting composition from the experiments of Blatter et al. (2013, 2023) is recommended, as it yields consistent results between experiments and MAGEMin calculations. The thermodynamic models exhibit difficulty in predicting the abundances of accessory phases, reflecting their limited contribution to the total Gibbs free energy of the system; such results should therefore be interpreted with caution.

The thermodynamic databases reproduce the evolution of major oxides SiO_2 , CaO , FeO , Al_2O_3 and MgO with high accuracy, whereas larger uncertainties are associated with Na_2O , K_2O , and TiO_2 . Overall, this accuracy is sufficient for modeling the thermodynamic evolution of silicate melts during magmatic differentiation, including key properties such as liquid fraction and density (Figure 7).

3.5. Enhancing Neural Network Performance Through Cross-Validation and Hyperparameter Tuning

At low batch sizes (Figure 8a), the validation loss shows a minimum toward high numbers of neurons and a low number of layers. Toward deeper architectures (i.e., higher number of layers), the validation loss increases at high-neuron numbers for the lowest tested batch size which might indicate overfitting. Increasing the batch size stabilizes the average validation loss field with an expected minimum for deep architectures (high number of neurons and high number of layers, Figures 8b and 8c). The surrogate model comprising three layers, 256 neurons, and trained on a batch size of 16,384 samples yielded the smallest mean absolute error (MAE) on the validation data (see white star in Figure 8f). This observation is consistent with the findings of Kerswell et al. (2024) who also report higher accuracy for deep neural network architectures.

The batch size is a hyperparameter that directly affects training time. In our experiments, the network trained with the largest tested batch size (16,384) achieved the best generalization. Although large batch sizes are often reported to harm generalization—because the optimization may converge to sharp minima (Keskar et al., 2017; LeCun et al., 2012)—this effect depends strongly on the data set and training setup. In many studies, networks trained with smaller batches outperform those using larger ones (Smith et al., 2020). Identifying the optimal batch size is therefore challenging, as it depends on several factors, including the optimizer, data properties, and network architecture (e.g., Shallue et al., 2019). Recent work by Zhang et al. (2019) and L. Chen et al. (2018) shows that combining the Adam optimizer with wider rather than deeper networks can increase the critical batch size—the point beyond which generalization begins to deteriorate. Our results support these findings: for thermodynamic data sets, a wide network with 256 neurons per layer and an intermediate depth of three layers outperformed narrower architectures trained with smaller batch sizes (Figure 8).

Throughout the training epochs, the validation loss consistently remains higher than the training loss and begins to plateau after approximately 475 epochs (Figure 8g). Losses decrease by two orders of magnitude. The sharp increase between 50 and 75 epochs can be attributed to the switch from the warm-up phase with constant learning rate to the true training phase in which the learning rate is reduced following cosine annealing.

3.6. Surrogate Model Accuracy

Figure 9 compares surrogate predictions of output parameters to the corresponding MAGEMin ground truth of a testing set. The parameters shown in Figures 9a–9m are direct surrogate output parameters. Figure 9n shows the solid density, which is calculated in post-processing using Equation 1. Overall, the surrogate-predicted values agree well with the true test data (all ca. 50,000 test data points fall approximately on the diagonal). As noted earlier, the denominator in Equation 1 approaches zero at high mobile phase fractions, leading to large errors in post-processed solid properties. By propagating the residual covariance of the surrogate model's predictions into the solid density calculation (Equation 1), we find that prediction errors exceed 10% of the absolute density value when the mobile phase fraction exceeds 80 wt.% (see Text S5 and Figure S5 in Supporting Information S1). Therefore, we recommend directly training the network on liquid, fluid, and solid properties when high mobile phase fractions are anticipated.

The uncertainties of predictions are represented by the MAE and by how much the difference between prediction and ground truth deviates from the MAE in Figure 10. Our surrogate model attains a classification accuracy of 98% for both the fluid and liquid classifier (true positives + true negatives in Figures 10c and 10f). Although such

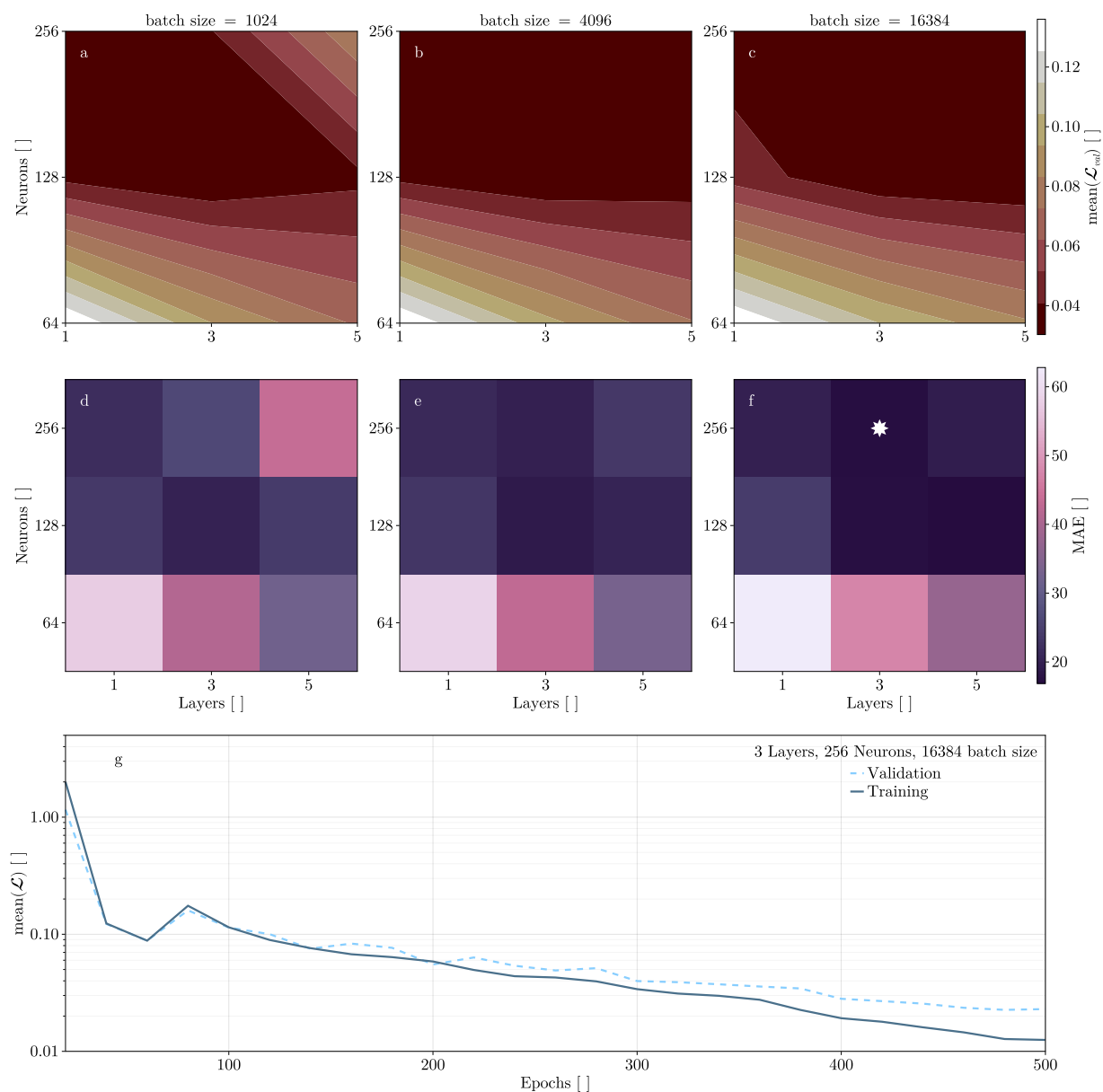


Figure 8. Cross validation results. Panels (a–c) show the validation loss averaged over the folds as maps over the number of layers and neurons for different batch sizes. Panels (d–f) show the summed mean absolute error of predicted outputs with the white star marking the lowest value, and (g) shows the evolution of the train and validation loss averaged over the folds for the surrogate model with the hyperparameters marked with the white star in (f).

values may initially appear implausibly high, they become attainable when the surrogate is interpolating in a very tight data range. A comparable ceiling of accuracy has also been observed in both geochemistry (Tan et al., 2025) and industrial studies on defect detection (Raveendran & Chandrasekhar, 2022). When extrapolating to conditions beyond the envelope of the training data, the accuracy will likely deteriorate and the user should reassess the predictive power of the surrogate model.

We report MAE and RMSE for the output parameters in Table 2. With a RMSE of ca. 16 kg m^{-3} the system density error is slightly lower compared to the error reported by Kerswell et al. (2024), but overall in the same range. This confirms that our more complex network architecture with branched classifier and regression heads maintains and potentially improves the accuracy of predictions compared to existing surrogate models. Such a design is appropriate when the compositional range of the data and dimensionality increases, as evidenced by the networks ability to predict the liquid major oxide concentrations with uncertainties below 1 wt.% for SiO_2 , Al_2O_3 ,

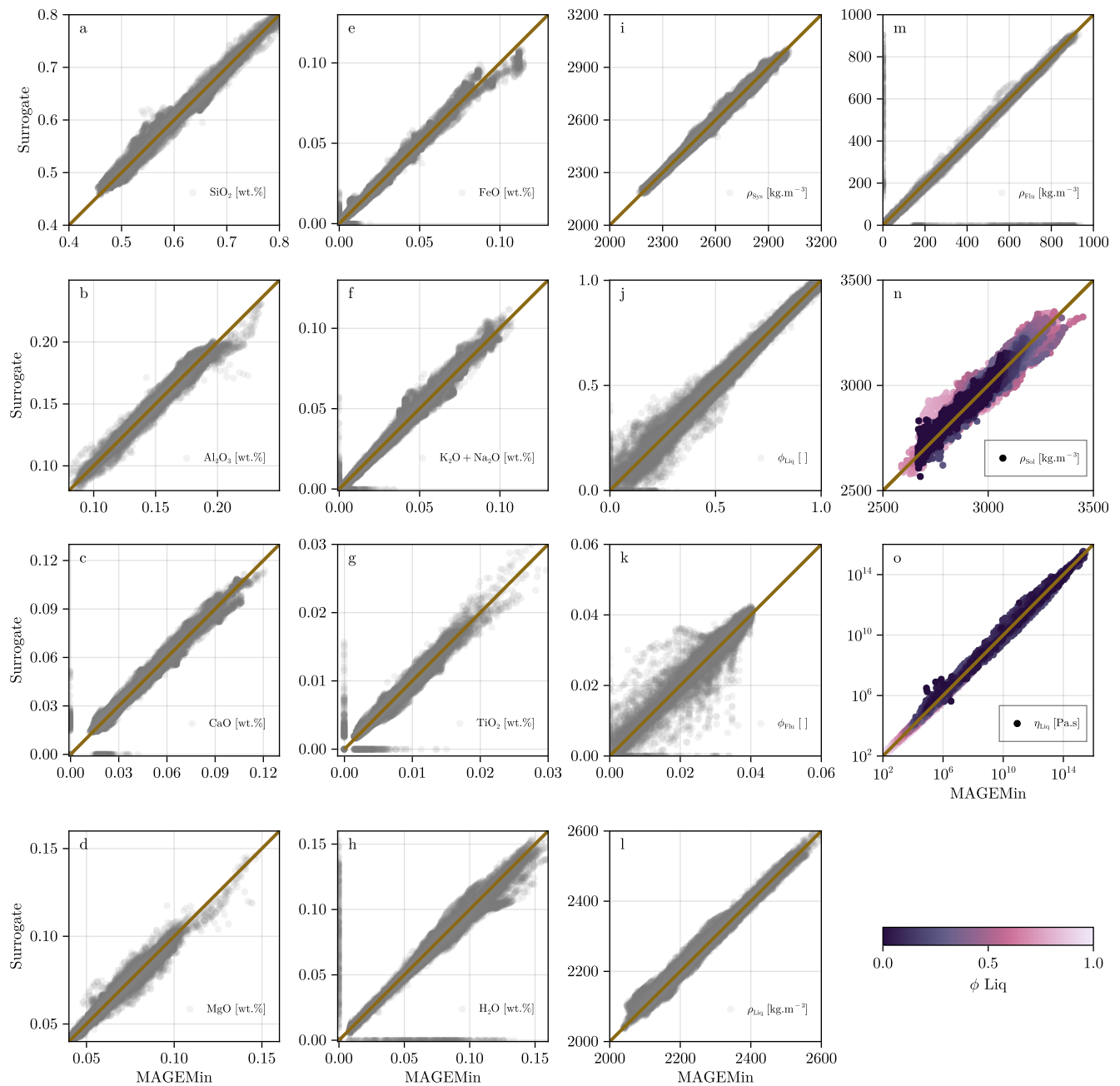


Figure 9. Accuracy of surrogate-model predictions compared to true values calculated by MAGEMin. We include true positives, false positives, true negative, false negative predictions. (a–h) Major oxide concentrations in the liquid, (i) system density, (j) liquid fraction, (k) fluid fraction, (l, m) liquid, and fluid density, (n) solid density, and (o) the liquid viscosity calculate after Giordano et al. (2008) (see Text S4 in Supporting Information S1 for details). The parameters shown in (a–m) are direct output parameters of the surrogate, (n–o) are calculated from these outputs in post-processing. The total number of test points: ca. 50,000. The solid golden lines indicate unity, that is MAGEMin and surrogate model predictions are identical.

and CaO , and below 0.5 wt.% for the other oxides (see Figures 10a and 10d). Furthermore, the composition uncertainties introduced by the surrogate model do not have a significant impact on the derived quantities such as the liquid viscosity (Figure 9o).

Uncertainties are generally larger for liquid properties than for fluid properties (see Figures 10b and 10e). This likely reflects (a) the larger variability in liquid data in comparison to fluid data and (b) the broader range of liquid fractions (0–100 wt.%) compared to fluid fractions (0–4 wt.%) which creates imbalanced losses and gradients during the training process, leading to larger errors even for the dominating quantity (W. Chen et al., 2024). In

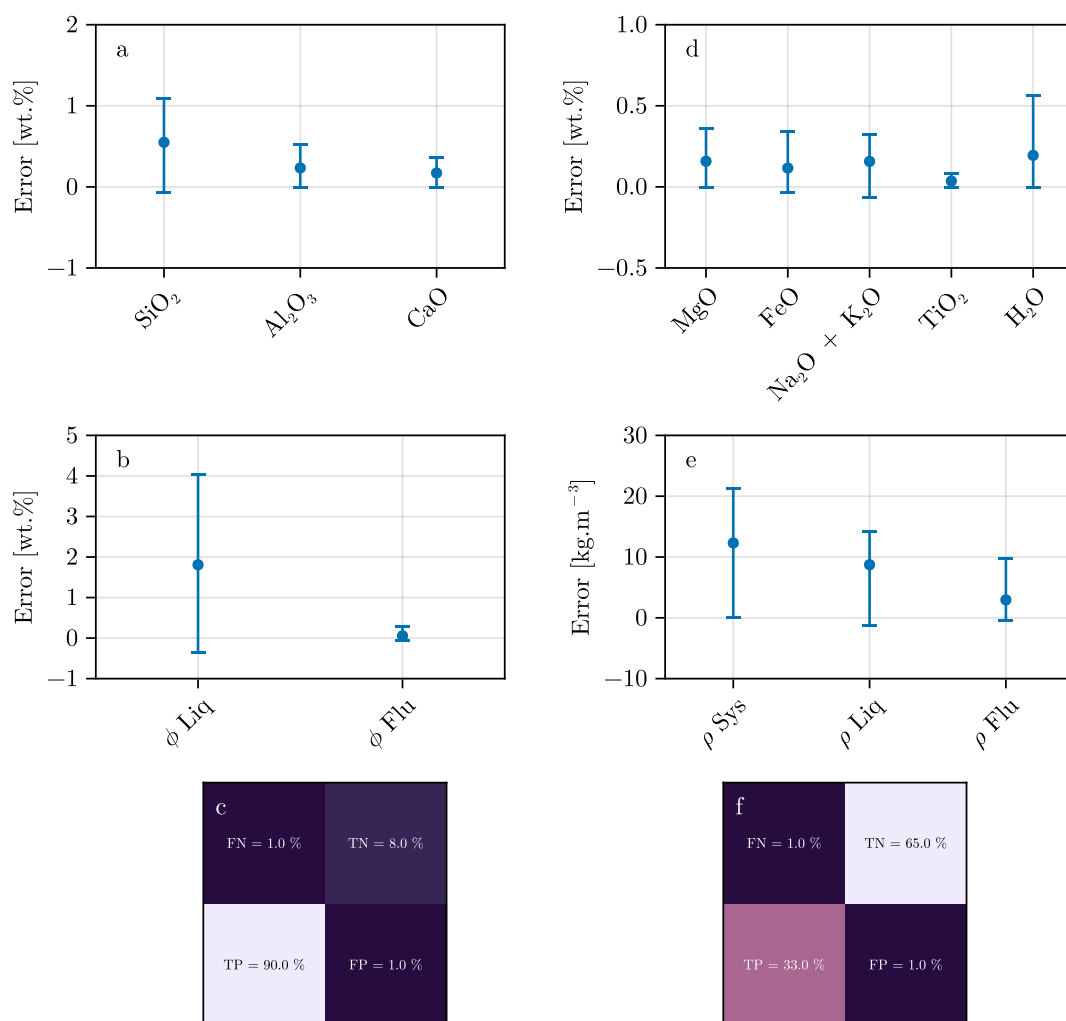


Figure 10. Uncertainty of surrogate model predictions. We only include true positive and true negative predictions. (a) Liquid concentration of SiO₂, Al₂O₃, and CaO, (b) liquid and fluid concentration, (c) confusion matrix for liquid classification, (d) liquid concentration of MgO, FeO, Na₂O + K₂O, TiO₂, and H₂O, (e) system, liquid, and fluid density, and (f) confusion matrix for fluid classification. In panels (a), (b), (d), and (e) the circle represents the arithmetic average error and bars denote the standard deviation. In panels (c) and (f), TP = true positive, FP = false positive, TN = true negative, FN = false negative. The percentage indicates the count relative to the total number of test data points (ca. 50,000).

most industrial applications, such imbalances are mitigated by either oversampling, that is duplicating, the data points of the imbalanced class, by scaling the gradients (W. Chen et al., 2024), or weighting the losses (Kendall et al., 2018). For fully synthetic data sets such as ours, however, the data range and size can be adjusted according to the compositions or liquid fractions expected during arc magma evolution. Aligning the training data more closely with the relevant parameter space, would reduce this numerical imbalance a priori and likely further improve the regression accuracy, without duplicating data or modifying gradients during the training process.

3.7. Rapid Inference of Thermodynamic Properties

We conducted a general benchmark to evaluate the computational performance of the surrogate relative to MAGEMin. Details of the benchmarking procedure are provided in the Text S3 in Supporting Information S1.

Using a single CPU thread for both MAGEMin and the surrogate model (iso-resource benchmark), the surrogate call is approximately 22,000× faster compared to the MAGEMin call once the problem size exceeds 10,000 data points (Figure 11a). When both methods are executed on 64 CPU-threads (peak-performance benchmark), the speed-up remains at about 3,400× (Figure 11c). This estimate is more realistic for applications where MAGEMin

Table 2
Mean Absolute Error (MAE), Root-Mean-Square Error (RMSE), and Standard Deviation of the Absolute Error of Surrogate-Model Predictions

Output parameter	MAE	RMSE	σ (AE)
SiO ₂ ^{Liq}	0.54	0.79	0.58
Al ₂ O ₃ ^{Liq}	0.23	0.33	0.24
CaO ^{Liq}	0.17	0.24	0.17
MgO ^{Liq}	0.15	0.23	0.17
FeO ^{Liq}	0.11	0.20	0.17
(Na ₂ O + K ₂ O) ^{Liq}	0.15	0.25	0.21
TiO ₂ ^{Liq}	0.03	0.05	0.04
H ₂ O ^{Liq}	0.19	0.33	0.27
ρ^{Sys}	12.30	16.10	10.39
ϕ^{Liq}	1.80	2.76	2.09
ϕ^{Flu}	0.06	0.15	0.14
ρ^{Liq}	8.66	12.03	8.61
ρ^{Flu}	2.95	6.71	6.03

Note. Values are calculated only for data points for which the classifier predictions are correct (i.e., true positives and true negatives). Units of major oxides, phase fractions, and densities are in wt.%, wt.%, and kg m⁻³, respectively.

would be run in parallel, such as in coupled geodynamic simulations. Compared to similar networks presented in previous studies, our surrogate model achieves 3–30× higher acceleration factors (Kerswell et al., 2024; Leal et al., 2020), underscoring its computational efficiency.

To ensure a fair comparison, we also evaluated the effect of providing MAGEMin with an initial guess from a previous solution; a built-in functionality of the software. Test compositions were generated as normally distributed random variations around this initial guess, with the degree of variation controlled by a standard deviation (σ in Figure 11). The closer a composition lies to the initial guess (i.e., smaller σ in Figures 11a and 11c), the faster the convergence, reducing execution time by ca. a factor of 2 in addition to MAGEMin's multi-threading parallelization (Figure 11c). In the iso-resource benchmark, the use of an initial guess yields speeds up MAGEMin calls by up to 10× highlighting the advantage of this initialization strategy for traditional thermodynamic solvers.

MAGEMin_C.jl serves as a Julia wrapper for the original C library of MAGEMin. Because memory of input and output variables to these wrappers must be allocated from Julia, the memory footprint for both MAGEMin and the CPU-executed surrogate model is substantial and increases with the number of points analyzed (see Figures 11b and 11d). Such growth can limit performance (e.g., Farhana et al., 2019). Previous studies have reported additional speed-ups of 10–20× when memory allocations are minimized (Bishnu et al., 2023), suggesting that reprogramming the inference routine to reuse pre-allocated memory could further accelerate both MAGEMin_C and surrogate-model evaluations.

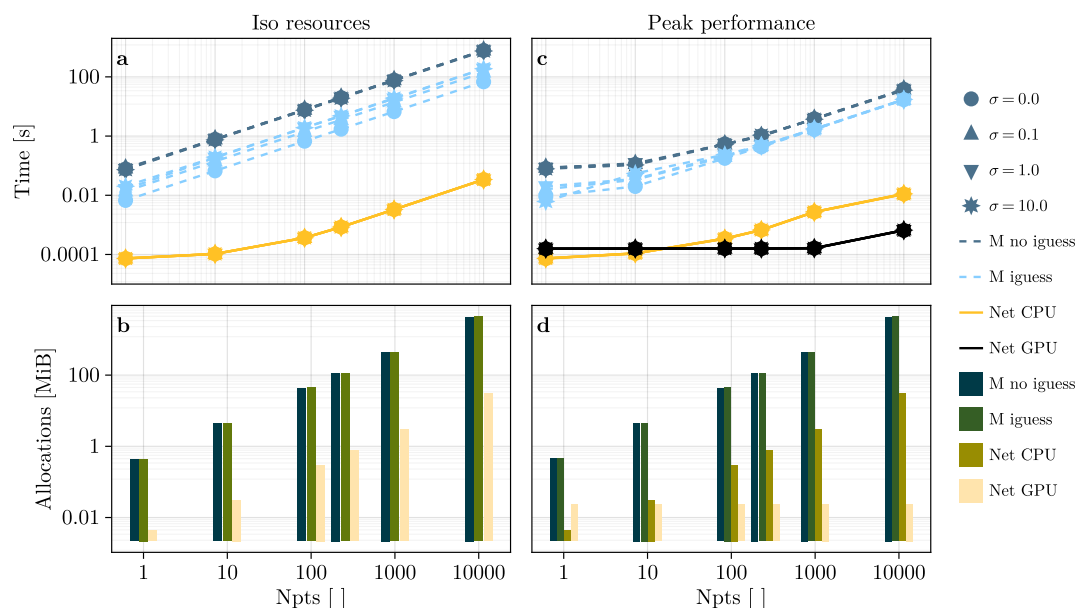


Figure 11. Time spent in, (a), and memory allocated, (b), by MAGEMin_C.jl's *multi_point_minimization()* (dark and light blue dashed lines) and Lux.jl's *apply()* function during the iso-resource benchmark. Panels (c) and (d) show the same quantities for the peak performance benchmark where MAGEMin (M) and surrogate model (Net) calls are multi-threaded and executed on the maximum available threads (64) in separate Julia sessions. The black line with star markers shows the timing for surrogate model calls on the graphics processing unit (GPU). Using an initial guess (with standard deviations, σ) speeds up MAGEMin computations by a factor of ca. 5. The surrogate call is a vectorized operation and up to 3,000 times faster compared to MAGEMin calculations. Memory allocations during MAGEMin and surrogate-model calls on the CPU increase with increasing number of data points. Surrogate calls on the GPU do not allocate significantly more memory as the number of data points increases.

Parallelization on the CPU does not significantly improve the speed of surrogate calls (compare yellow lines in Figures 11a and 11c), indicating that Matrix-Vector operations benefit little from multi-threading. In contrast to MAGEMin or other GFEM packages, these operations can exploit GPU-acceleration. Figure 11c shows that GPU execution times remain below 1 ms even when evaluating 10,000 data points simultaneously. Such performance is well within the range required for coupling with geodynamic codes (Kerswell et al., 2024). Moreover, the GPU memory footprint remains nearly constant with an increasing number of data points evaluated, an essential property to prevent performance degradation (Farhana et al., 2019).

3.8. Coupling the Surrogate Model to Geodynamic Codes

Only a few studies have solved GFEM and transport equations simultaneously during the runtime of geodynamic simulations (e.g., Tirone et al., 2009). More commonly, thermodynamic software is used to precompute a lookup table of properties across the relevant P–T–X space for studies of multiphase reactive flow (e.g., Bessat et al., 2022; Katz, 2008; Rummel, Kaus, et al., 2020). During the geodynamic calculation, these precomputed quantities required by the transport algorithm are interpolated according to the local grid-cell conditions.

Most modern geodynamic algorithms are based on theories that are only valid within specific limits. At low liquid fractions, in the porous flow regime, the small porosity approximation (e.g., Keller & Katz, 2016; McKenzie, 1989) is widely used to simulate mantle-scale reactive flow. As the liquid fraction increases and magma ascends into the lithosphere forming reservoirs, this approximation breaks down because the shear stress inside the pore liquid cannot be neglected. At high liquid fractions, in the suspension flow regime, the solid is only present in form of individual crystals that settle according to Stokes' law (e.g., Dufek & Bachmann, 2010). Recently, Keller and Suckale (2019) developed a unified theory for multi-phase transport that is valid across the different limits and Wong and Keller (2023) subsequently tested geodynamic models based on this new theory. However, because these processes operate on different spatial and temporal scales, the models require different numerical resolutions. Despite the unified theory, it remains more practical to study the different reactive transport end-members in separate numerical simulations, as demonstrated by Wong and Keller (2023).

For these reasons, a single, general surrogate model that interpolates thermodynamic data ranging across the different end-members is impractical. Instead, coupling between geodynamic and surrogate models should be tested separately for each end-member case in a systematic benchmarking framework which is beyond the scope of this study. The surrogate model presented here should, therefore, be regarded as a proof of concept for rapid inference of thermodynamic properties. Nevertheless, several considerations arise for developing surrogate models intended for coupling with geodynamic codes.

1. *Training data scope.* The data set should be carefully selected and restricted to the parameter range relevant for the specific end-member. For example, in the small-porosity limit, data for liquid fractions above a few wt.% are unnecessary. This narrowing of range reduces the numerical gap between fluid and liquid quantities and likely improves the prediction of liquid properties.
2. *Choice of outputs.* The predicted thermodynamic parameters should be consistent with the thermodynamic admissibility of the reactive-transport formulation. Depending on the underlying geodynamic theory, the surrogate model may need to output quantities different from those presented here.
3. *Integration and performance.* To minimize performance losses associated with additional memory allocation, the inference routine should be implemented directly within the geodynamic framework to enable memory reuse.

4. Conclusion

We assessed the capability of MAGEMin, based on the thermodynamic data sets of Holland et al. (2018) and Green et al. (2025), to reproduce fractional crystallization experiments. The general agreement between thermodynamic-model predictions and laboratory data, including the accurate reproduction of key mineral phases, confirms that this thermodynamic formulation is suitable for modeling the evolution of arc magmas. However, the performance depends on the starting composition, and simulations should be restricted to well-benchmarked bulk compositions.

Using this validated thermodynamic model, we generated a synthetic data set covering primitive basaltic to rhyolitic compositions and temperature–pressure conditions relevant to the upper to lower crust. This data set was

used to train a neural network capable of interpolating thermodynamic properties within the trained compositional and P–T range. The resulting surrogate model correctly predicts liquid and fluid presence or absence with 98% accuracy and interpolates continuous thermodynamic properties with an average RMSE below 1 wt.% for compositions, below 20 kg m^{−3} for densities, and below 3 wt.% for phase fractions. Inference of a single data point takes approximately 0.1 ms, corresponding to a speed-up of about 1,000× compared to a direct MAGEMin call.

These results demonstrate that neural networks provide an efficient and accurate alternative to traditional lookup-table approaches for coupling thermodynamic calculations with transport or geodynamic models, substantially reducing computational cost while retaining predictive fidelity for compositionally evolving magmatic systems.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

All scripts to (a) run the comparison of thermodynamic models to laboratory experiments, (b) generate the training and testing data, (c) perform the hyperparameter study within the framework of cross-validation, (d) train and test the neural network, and (e) produce the data figures from the main article are available within the GitHub repository by L. Candioti (2026) archived on Zenodo (<https://doi.org/10.5281/zenodo.18770983>). This repository also hosts the table containing the compilation of laboratory experiments. The data on which the conclusions of this study are based is published in L. G. Candioti (2025) and archived on Zenodo (<https://doi.org/10.5281/zenodo.17369875>) as well.

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